

Evaluating Retrieval Robustness in RAG Systems Under Increasing Distractor Noise

Research Area:

Retrieval-Augmented Generation (RAG), Information Retrieval, NLP

Experimental Focus:

Robustness of BM25, Dense, and Hybrid Retrieval under Increasing Distractor Noise

Research 1

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Abstract

Retrieval quality is a critical component of Retrieval-Augmented Generation (RAG) systems because the documents selected by the retrieval stage determine the information available to downstream answer generation. This study investigates how increasing distractor-document noise affects retrieval stability, generated-answer quality, and latency across three retrieval approaches: BM25, dense retrieval, and hybrid retrieval.

The experiment evaluates 20 questions across four noise conditions: 0, 1, 2, and 4 injected distractor documents, resulting in 240 observations across the three retrieval methods. For each condition, the final retrieval output was limited to the top five documents. Retrieval stability was evaluated by measuring injected-distractor contamination and preservation of the baseline top-five document set. Generated answers were evaluated using Token F1 and semantic similarity, while latency was measured in seconds. Paired Wilcoxon signed-rank tests with Holm correction were used to compare each noisy condition against its corresponding baseline.

Across all 240 observations, none of the injected distractors entered the final top-five results, and all five baseline documents were preserved under every tested condition. No statistically significant changes in Token F1 or semantic similarity remained after multiple-comparison correction. The only statistically significant effect was observed for dense-retrieval latency at the

highest noise level, where mean latency increased from 0.4137 seconds to 0.6246 seconds, representing a 50.99% increase (Holm-adjusted $p = 0.0130$; rank-biserial correlation = 0.8286).

These results indicate that, under the specific corpus, retrieval configuration, distractor-generation procedure, and noise levels evaluated, the tested retrieval methods maintained stable top-five rankings despite increasing distractor injection. The findings also suggest that increased candidate-set complexity may introduce computational overhead even when retrieval rankings remain unchanged.

1. Introduction

Retrieval-Augmented Generation (RAG) systems combine information retrieval with language generation to produce answers grounded in a collection of external documents. In a typical RAG pipeline, a user query is first processed by a retrieval component that identifies a small set of potentially relevant documents. These documents are then provided as context to a language model, which generates the final response. Consequently, the quality and stability of the retrieval stage can directly affect the information available to the generation stage.

In practical retrieval systems, the candidate document pool may contain irrelevant or distracting information. As the number of distractor documents increases, a retrieval system must distinguish relevant documents from increasingly complex background information while maintaining an appropriate ranking of candidates. This creates an important robustness question: whether the addition of irrelevant documents changes the documents selected for the final context and, consequently, affects downstream answer generation.

Different retrieval approaches use different mechanisms for identifying relevant documents. BM25 is a sparse lexical retrieval method that ranks documents according to term-matching statistics. Dense retrieval represents queries and documents as vectors and measures their semantic similarity. Hybrid retrieval combines complementary lexical and semantic retrieval

signals. Comparing these approaches under controlled increases in distractor noise provides a way to examine whether their retrieval behavior remains stable as the candidate environment becomes more complex.

This study evaluates the behavior of BM25, dense, and hybrid retrieval under increasing distractor-document noise within a RAG pipeline. The experiment uses four noise conditions, ranging from no injected distractors to four distractor documents per query. Retrieval stability, answer quality, and latency are evaluated using controlled comparisons against the corresponding noise-free baseline.

The study is designed to distinguish between three related effects. First, distractor documents may enter the final retrieval results, directly changing the context presented to the generator. Second, changes in retrieved context may affect generated-answer quality. Third, even when the retrieved documents remain unchanged, processing a larger candidate set may increase computational latency. Examining these effects separately allows the experiment to determine which aspects of system behavior change under the tested noise conditions.

2. Research Question

The primary research question investigated in this study is:

RQ1: How does increasing distractor-document noise affect retrieval stability, generated-answer quality, and latency in BM25, dense, and hybrid retrieval systems within a RAG pipeline?

The study examines these effects by comparing each noisy retrieval condition with its corresponding noise-free baseline for the same question and retrieval method.

3. Hypotheses

Based on the research question, the study evaluates the following hypotheses:

H1 — Retrieval Stability: Increasing distractor-document noise will reduce retrieval stability, resulting in changes to the documents selected in the final top-five retrieval results.

H2 — Answer Quality: Increasing distractor-document noise will negatively affect generated-answer quality, as measured by Token F1 and semantic similarity.

H3 — Latency: Increasing distractor-document noise will increase retrieval and processing latency as the candidate document set becomes more complex.

These hypotheses are evaluated empirically using paired comparisons between each noise condition and its corresponding noise-free baseline.

4. Methodology

4.1 Experimental Design

The experiment was designed as a controlled evaluation of retrieval behavior under increasing distractor-document noise. Three retrieval approaches were evaluated: BM25, dense retrieval, and hybrid retrieval.

The experiment used 20 questions and four noise conditions. The baseline condition contained no injected distractor documents, while the remaining conditions introduced 1, 2, or 4 distractor documents. This produced noise ratios of 0.0, 0.1667, 0.2857, and 0.4444, respectively.

For every question, retrieval was performed with a fixed top-k value of 5. Each retrieval method was evaluated under every noise condition, producing 20 observations per method-condition combination. Across three retrieval methods and four noise levels, the complete experiment contained:

- **20 questions**
- **3 retrieval methods**
- **4 noise conditions**
- **240 total observations**
- **5 retrieved documents per observation**

The noise-free retrieval result for each question and method served as the baseline against which the corresponding noisy conditions were compared. This paired design controls for differences between questions when evaluating the effect of distractor injection.

The experiment evaluates three principal dimensions of system behavior: retrieval stability, generated-answer quality, and latency. Retrieval stability is assessed by determining whether injected distractors enter the final top-five results and whether the baseline top-five document set is preserved. Answer quality is evaluated using Token F1 and semantic similarity, while latency is measured in seconds.

Table 1. Experimental configuration

Parameter

Configuration

Evaluation questions	20
Retrieval methods	BM25, Dense, Hybrid
Noise levels	0, 1, 2, 4 distractors
Maximum noise ratio	0.4444
Top-k	5
Observations per method × noise condition	20
Total observations	240
Distractor sampling seed	42
Baseline	0 injected distractors
Statistical test	Paired Wilcoxon signed-rank
Multiple-comparison correction	Holm correction
Effect size	Rank-biserial correlation

4.2 Dataset

The experiment was conducted using a fixed evaluation set containing 20 questions. Each question was associated with a reference answer and a collection of documents used by the retrieval pipeline.

The same questions and underlying document collection were used across all retrieval methods and noise conditions to ensure controlled comparisons. For each experimental observation, the retrieval system produced a ranked set of five documents, which were subsequently used as context for answer generation.

The complete experimental dataset contained 240 observations: 20 questions evaluated across three retrieval methods and four noise conditions. The dataset recorded the question identifier, question text, reference answer, retrieval method, number of injected distractors, noise ratio, retrieved document identifiers, distractor identifiers, generated answer, latency, Token F1, and semantic similarity.

The fixed evaluation design ensures that differences between conditions can be attributed to the retrieval configuration and distractor manipulation rather than changes in the evaluated question set.

4.3 Retrieval Methods

Three retrieval approaches were evaluated in the experiment.

BM25: BM25 is a sparse lexical retrieval method that ranks documents according to the occurrence and importance of query terms within the document collection. It provides a term-based baseline for evaluating retrieval behavior.

Dense Retrieval: Dense retrieval represents queries and documents using vector embeddings and ranks documents according to their semantic similarity to the query. This allows retrieval based on semantic relationships rather than exact lexical overlap.

Hybrid Retrieval: Hybrid retrieval combines lexical and semantic retrieval signals. It was included to evaluate whether combining the two retrieval paradigms produces different behavior under increasing distractor noise.

All three methods were evaluated using the same questions, document collection, noise conditions, and top-five retrieval limit.

4.4 Distractor Injection

Distractor documents were introduced into the retrieval candidate environment to simulate increasing levels of irrelevant information. Four experimental noise levels were evaluated: 0, 1, 2, and 4 injected distractor documents.

The noise-free condition, with zero injected distractors, served as the baseline. The remaining conditions progressively increased the number of distractor documents available to the retrieval process. The corresponding noise ratios were 0.0, 0.1667, 0.2857, and 0.4444.

For each question and retrieval method, the noisy retrieval result was compared with its matched noise-free baseline. This allowed the experiment to determine whether the injected documents altered the final top-five ranking.

An important distinction was maintained between the expanded candidate context and the final retrieved result. The candidate environment could contain the original documents together with injected distractors, while the retrieval output was restricted to the top five documents. Distractor contamination was therefore defined specifically as the presence of an injected distractor within this final top-five result.

The experiment evaluated distractor behavior under the specific distractor-generation procedure used in the study. The results therefore describe robustness within the tested experimental conditions rather than universal resistance to arbitrary or adversarial distractors.

4.5 Evaluation Metrics

The experiment evaluated retrieval behavior and downstream generation using four primary measures: distractor contamination, baseline top-five preservation, Token F1, semantic similarity, and latency.

Distractor Contamination: Distractor contamination measures the proportion of final top-five retrieval results that consist of injected distractor documents. A value of 0 indicates that no injected distractors entered the final retrieved set.

Baseline Top-Five Preservation: Baseline preservation measures how much of the noise-free top-five retrieval set is retained under each noisy condition for the same question and retrieval method. A value of 1.0 indicates that all five documents present in the corresponding baseline result were preserved.

Token F1: Token F1 measures lexical overlap between the generated answer and the reference answer using precision and recall at the token level. Higher values indicate greater token-level agreement with the reference answer.

Semantic Similarity: Semantic similarity measures the semantic correspondence between the generated answer and the reference answer. Higher values indicate greater semantic similarity.

Latency: Latency measures the time required for the evaluated retrieval and generation operation, recorded in seconds. Changes in latency were examined to determine whether increasing distractor complexity introduced computational overhead.

4.6 Statistical Analysis

Each non-baseline noise condition was compared with the corresponding noise-free baseline for the same question and retrieval method. This produced paired observations for each combination of retrieval method, noise level, and evaluation metric.

The paired Wilcoxon signed-rank test was used to evaluate whether the distribution of each metric differed between a noisy condition and its matched baseline. This non-parametric test was selected for the paired comparison because it does not require the differences to follow a normal distribution.

The analysis included three retrieval methods, three non-zero noise levels, and three evaluated metrics, resulting in 27 paired statistical comparisons.

Because multiple statistical tests were performed, p-values were adjusted using the Holm correction to control the family-wise error rate. Statistical significance was evaluated using the Holm-adjusted p-value rather than the raw p-value.

Rank-biserial correlation was also calculated as an effect-size measure for the paired comparisons. This provides an indication of the direction and magnitude of the observed paired difference in addition to statistical significance.

The noise-free condition was treated as the reference baseline throughout the statistical analysis. No statistical comparison was performed between different retrieval methods because the primary objective was to measure the effect of increasing distractor noise within each method.

5. Results

The experiment produced 240 observations across 20 questions, three retrieval methods, and four noise conditions. The results are organized into four areas: retrieval stability, generated-answer quality, latency, and statistical significance.

5.1 Retrieval Stability

Across all 240 observations, none of the injected distractor documents entered the final top-five retrieval results. The observed distractor contamination rate was therefore **0.0** across every retrieval method and noise condition.

The baseline top-five retrieval set was also completely preserved in every noisy condition. Baseline top-five preservation was **1.0** for BM25, dense retrieval, and hybrid retrieval at noise levels 1, 2, and 4.

The maximum evaluated condition contained four injected distractors, corresponding to a noise ratio of **0.4444**. Even at this highest tested noise level, all five documents from the corresponding noise-free baseline remained in the final top-five result.

These observations indicate that, within the evaluated experimental configuration, the injected distractors did not alter the final top-five document sets.

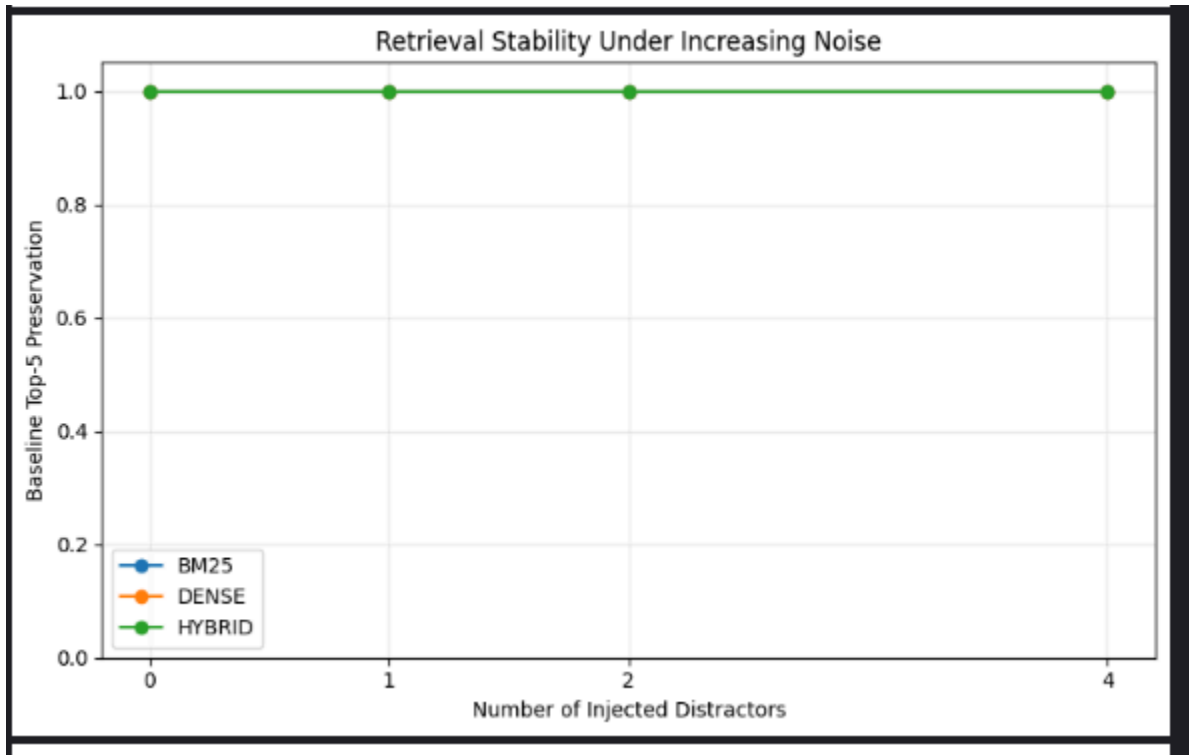
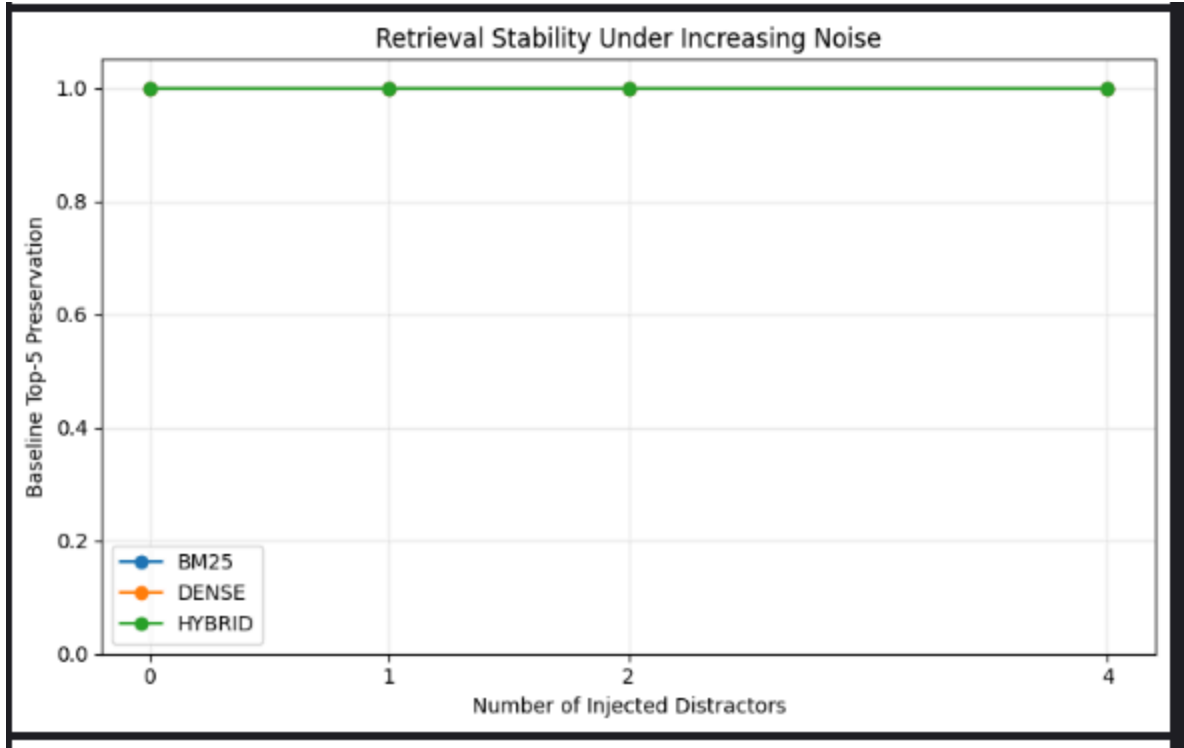


Table 2. Retrieval stability across noise conditions

Retrieval Method	Noise Level	Noise Ratio	Distractor Contamination	Baseline Top-5 Preservation	Observations
BM25	0	0.0000	0.0000	1.0000	20

BM25	1	0.1667	0.0000	1.0000	20
BM25	2	0.2857	0.0000	1.0000	20
BM25	4	0.4444	0.0000	1.0000	20
Dense	0	0.0000	0.0000	1.0000	20
Dense	1	0.1667	0.0000	1.0000	20
Dense	2	0.2857	0.0000	1.0000	20
Dense	4	0.4444	0.0000	1.0000	20
Hybrid	0	0.0000	0.0000	1.0000	20
Hybrid	1	0.1667	0.0000	1.0000	20
Hybrid	2	0.2857	0.0000	1.0000	20
Hybrid	4	0.4444	0.0000	1.0000	20

Note. Distractor contamination measures the proportion of final top-five retrieved documents that were injected distractors. Baseline top-5 preservation measures the proportion of the corresponding noise-free top-five set retained under each noisy condition.



5.2 Answer Quality

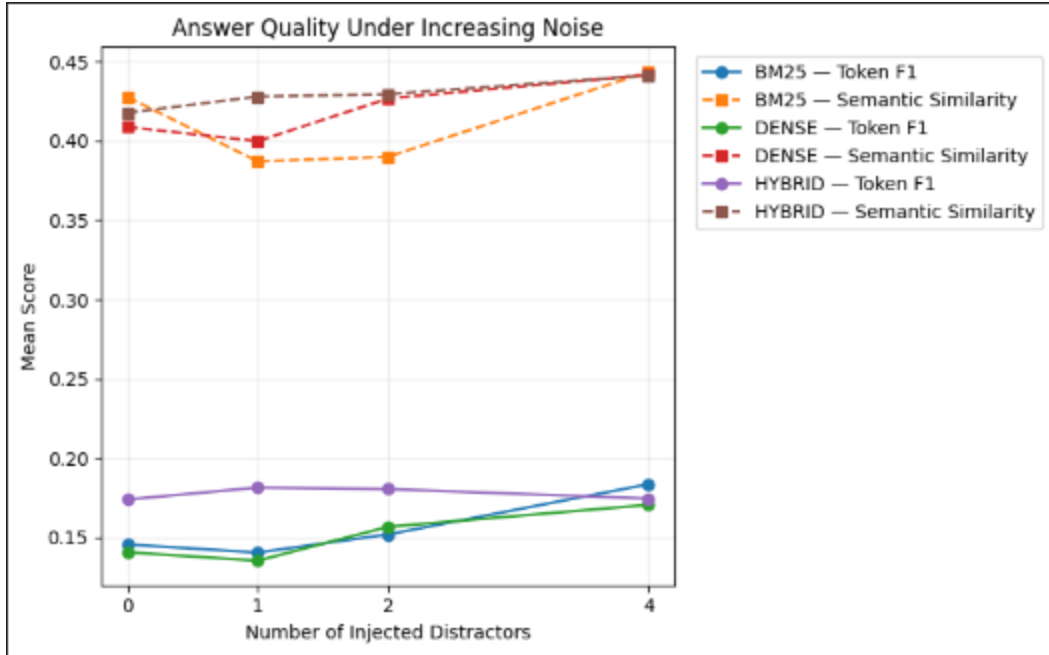
Table 3. Mean answer-quality metrics across noise conditions

Retrieval Method	Noise	Token F1	Semantic Similarity
BM25	0	0.1482	0.4076
BM25	1	0.1348	0.3855
BM25	2	0.1554	0.4087
BM25	4	0.1791	0.4442

Dense	0	0.1452	0.4127
Dense	1	0.1298	0.3945
Dense	2	0.1467	0.4153
Dense	4	0.1752	0.4687
Hybrid	0	0.1693	0.4538
Hybrid	1	0.1768	0.4426
Hybrid	2	0.1796	0.4214
Hybrid	4	0.1683	0.4384

Note. Values represent means across the 20 questions in each method × noise condition.

The descriptive results do not show a consistent monotonic decrease in answer quality as the number of distractors increased. Because descriptive differences alone do not establish statistical significance, paired statistical testing with Holm correction was used to determine whether these observed changes were statistically supported. No Token F1 or semantic-similarity comparison remained significant after correction.



5.3 Latency

Latency was measured in seconds for each experimental observation. Mean latency generally increased relative to the noise-free baseline as the number of distractor documents increased, although the magnitude of change differed across retrieval methods.

For BM25, mean latency increased from 0.4379 seconds at baseline to 0.5666 seconds with one distractor, 0.5085 seconds with two distractors, and 0.5330 seconds with four distractors.

For dense retrieval, mean latency increased from 0.4137 seconds at baseline to 0.4542 seconds with one distractor, 0.5121 seconds with two distractors, and 0.6246 seconds with four distractors.

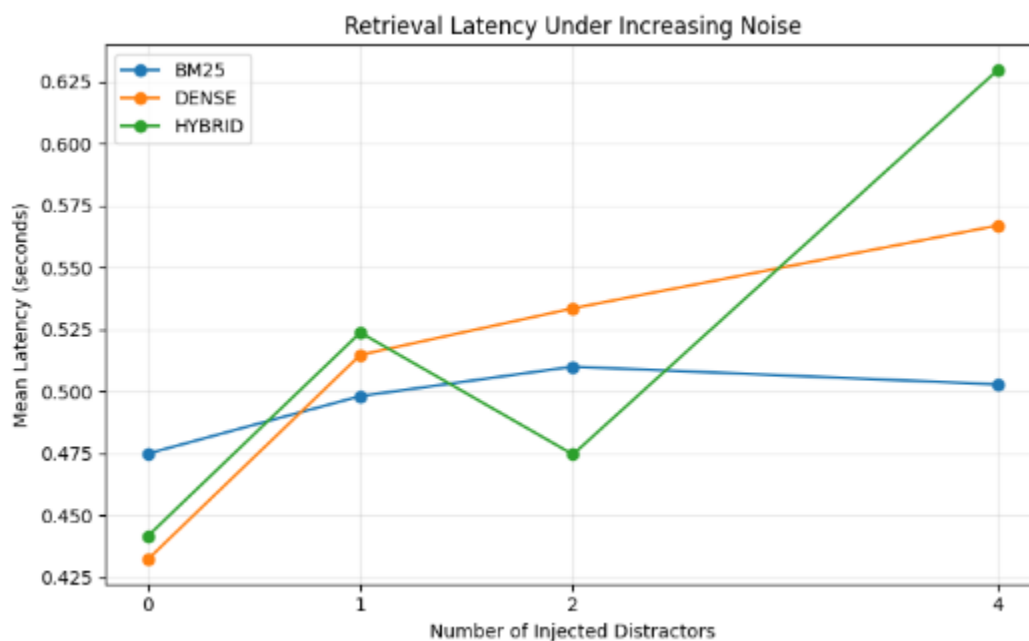
For hybrid retrieval, mean latency increased from 0.4662 seconds at baseline to 0.4765 seconds with one distractor, 0.4919 seconds with two distractors, and 0.6025 seconds with four distractors.

The largest descriptive increase occurred for dense retrieval at the highest noise condition, where mean latency increased by 0.2109 seconds compared with the baseline.

Table 4. Paired statistical comparison surviving Holm correction

Retrieval Method	Noise Level	Metric	Baseline Mean	Noisy Mean	Absolute Change	Relative Change	Raw p	Holm-adjusted p	Rank-Biserial
Dense	4	Latency	0.4137	0.6246 s	+0.2109 s	+50.99 %	0.0005	0.0130	0.8286

Note. A total of 27 paired comparisons were performed. This table reports the only comparison that remained statistically significant after Holm correction at $\alpha = 0.05$.



5.4 Statistical Results

A total of 27 paired statistical comparisons were conducted across the three retrieval methods, three non-zero noise levels, and three evaluation metrics. P-values were adjusted using the Holm correction to account for multiple comparisons.

After Holm correction, **one of the 27 comparisons remained statistically significant**. This comparison involved dense retrieval latency at the highest noise condition of four injected distractors.

For dense retrieval, mean latency increased from **0.4137 seconds** at baseline to **0.6246 seconds** under four-distractor noise, corresponding to a **50.99% increase**. The paired Wilcoxon signed-rank test produced a raw p-value of approximately **0.0005**, with a Holm-adjusted p-value of **0.0130**. The rank-biserial correlation was **0.8286**.

No Token F1 or semantic similarity comparison remained statistically significant after Holm correction. Therefore, the statistical analysis provides no evidence of a systematic answer-quality change attributable to the tested distractor conditions after correction for multiple comparisons.

The statistical results should be interpreted together with the retrieval-stability results: because no injected distractor entered the final top-five retrieval results, the experiment did not observe retrieval-context contamination under the tested distractor construction.

Table 5 .statistical result

Retrieval Method	Noise 0	Noise 1	Noise 2	Noise 4	Change at Noise 4
BM25	0.4379 s	0.5666 s	0.5085 s	0.5330 s	+21.72%
Dense	0.4137 s	0.4542 s	0.5121 s	0.6246 s	+50.99%
Hybrid	0.4662 s	0.4765 s	0.4919 s	0.6025 s	+29.25%

6. Discussion

The experimental results provide a mixed picture of RAG behavior under increasing distractor noise. Across the evaluated corpus, retrieval configuration, and distractor-generation procedure, BM25, dense, and hybrid retrieval maintained the same top-5 document sets at all tested noise levels. No injected distractor entered the final top-5 results in any of the 240 experimental

observations. Consequently, distractor contamination remained 0.0 and baseline top-5 preservation remained 1.0 throughout the experiment.

These results provide evidence of strong retrieval stability under the specific experimental conditions used in this study. However, they should not be interpreted as evidence that the evaluated retrieval methods are universally robust to distractor noise. The experiment tested a controlled form of noise in which distractors were sampled from documents outside the method's baseline top-5 results. Different distractor-generation strategies, larger corpora, higher noise levels, or semantically harder distractors could produce different outcomes.

6.1 Retrieval Stability

The retrieval-stability results support the observation that the injected distractors did not alter the ranking of the five documents selected by any retrieval method. This is particularly notable because the experiment increased the number of injected distractors from zero to four.

However, the absence of top-k contamination also affects how the answer-quality results should be interpreted. Since the retrieved document sets remained unchanged, changes in generated-answer quality cannot be directly attributed to distractors replacing relevant documents in the retrieved context.

The results therefore suggest that, under this experimental setup, retrieval ranking was more stable than answer-quality metrics alone might indicate.

6.2 Answer Quality

Token-level F1 and semantic similarity did not show a consistent monotonic degradation as noise increased. For example, BM25 showed lower mean F1 and semantic similarity at noise level 1, followed by higher values at noise levels 2 and 4. Dense retrieval displayed a similar non-monotonic pattern, while hybrid retrieval showed smaller variations across the tested conditions.

Because the injected distractors did not enter the top-5 retrieved set, these variations should not be interpreted as evidence that distractor documents directly degraded answer quality. Instead, they indicate that generated-answer metrics can vary across experimental conditions even when the retrieved document set remains unchanged.

The results therefore do not provide strong evidence for a systematic negative effect of increasing distractor noise on answer quality under the tested configuration.

6.3 Latency

Latency showed the clearest measurable effect. All three retrieval methods exhibited higher mean latency at noise level 4 than at the baseline condition.

BM25 increased from 0.4379 seconds to 0.5330 seconds, representing a 21.72% increase. Dense retrieval increased from 0.4137 seconds to 0.6246 seconds, representing a 50.99% increase. Hybrid retrieval increased from 0.4662 seconds to 0.6025 seconds, representing a 29.25% increase.

The paired statistical analysis identified the dense-retrieval noise-4 latency comparison as the only comparison that remained statistically significant after Holm correction. The mean latency increased by approximately 0.2109 seconds, with a Holm-adjusted p-value of 0.0130 and a rank-biserial correlation of 0.8286.

This result provides the strongest evidence in the experiment that increasing the evaluated noise condition was associated with increased computational latency, particularly for dense retrieval.

6.4 Interpretation of the Hypotheses

H1 — Retrieval stability: The hypothesis predicted that increasing distractor noise would reduce retrieval stability. This was not supported under the tested configuration because no distractor entered the top-5 results and baseline top-5 preservation remained 100% across all conditions.

H2 — Answer quality: The hypothesis predicted that increasing distractor noise would negatively affect generated-answer quality. The results did not provide statistically significant evidence for this effect after Holm correction. Answer-quality metrics also exhibited non-monotonic changes rather than a consistent decline.

H3 — Latency: The hypothesis predicted that increasing distractor noise would increase computational latency. The results provide partial support for this hypothesis. Latency increased at the highest tested noise level for all three methods, and the dense-retrieval noise-4 comparison remained statistically significant after multiple-comparison correction.

Overall, the experiment provides the strongest empirical evidence for a latency effect rather than a retrieval-ranking or answer-quality degradation effect.

7. Limitations

Several limitations constrain the interpretation and generalizability of these findings.

First, the experiment used 20 evaluation questions. Although the paired experimental design provides repeated measurements across retrieval methods and noise levels, a larger evaluation set would provide broader coverage of query types and reduce dependence on the characteristics of the selected questions.

Second, only three retrieval strategies were evaluated: BM25, dense retrieval, and a hybrid combination of the two. Other retrieval architectures, reranking models, multi-stage retrieval pipelines, and learned sparse retrieval methods may behave differently.

Third, the experiment evaluated only four noise levels: 0, 1, 2, and 4 distractors. Higher distractor counts and substantially larger noise ratios may reveal failure modes that were not observable in the tested range.

Fourth, distractors were sampled from documents outside the baseline top-5 results. This creates a controlled noise condition but does not necessarily represent the most difficult form of retrieval interference. Future experiments should include semantically similar distractors, adversarial distractors, near-duplicate documents, and documents containing misleading but plausible information.

Fifth, the absence of distractor contamination means that the experiment does not isolate the effect of incorrect documents entering the final context. A separate experiment in which distractors are explicitly designed to compete with relevant documents would be necessary to evaluate that failure mode.

Finally, the latency measurements are environment-dependent. Hardware, implementation details, model loading, network conditions, caching, and service availability can affect absolute latency. Therefore, the latency findings should primarily be interpreted within the experimental environment rather than as universal timing estimates.

8. Conclusion

This study evaluated the behavior of BM25, dense, and hybrid retrieval within a RAG pipeline under increasing distractor noise. Across 240 experimental observations, none of the injected distractors entered the final top-5 retrieval results, and all five baseline documents were preserved under every tested noise condition.

The experiment therefore did not observe a measurable degradation in retrieval stability under the controlled distractor-generation procedure. Similarly, token-level F1 and semantic similarity did not demonstrate a statistically significant degradation after Holm correction. Their changes were non-monotonic and cannot be directly attributed to distractor contamination because the retrieved document sets remained unchanged.

Latency produced the clearest effect. At the highest tested noise level, mean latency increased for BM25, dense, and hybrid retrieval. The dense-retrieval increase from 0.4137 seconds to 0.6246 seconds was statistically significant after Holm correction, with a 50.99% relative increase.

The main finding is therefore not that RAG systems are universally robust to noise, but that the evaluated retrieval configurations maintained ranking stability under the specific corpus, distractor construction, noise levels, and experimental procedure used in this study, while increased noise was associated with measurable latency overhead.

Future work should expand the evaluation set, increase noise levels, introduce semantically harder distractors, and explicitly test cases where distractors compete with relevant documents for top-k positions. Such experiments would provide a stronger assessment of retrieval failure modes and the robustness limits of RAG systems.

9. Reproducibility

The experiment was implemented as a controlled computational study with deterministic distractor sampling.

The experimental configuration consisted of:

- **Evaluation questions:** 20
- **Retrieval methods:** BM25, dense, hybrid
- **Noise levels:** 0, 1, 2, and 4 distractors
- **Top-k:** 5
- **Total observations:** 240
- **Random seed:** 42
- **Baseline:** zero injected distractors
- **Statistical test:** paired Wilcoxon signed-rank test
- **Multiple-comparison correction:** Holm correction
- **Effect size:** rank-biserial correlation

For each question and retrieval method, the baseline top-5 results were first obtained. Distractors were then sampled from documents that were not part of the baseline retrieved set. The same question-method pair was used as the baseline for each corresponding noisy condition.

The complete experiment produced the following primary artifacts:

1. Retrieval and generation results CSV
2. Descriptive experimental summary
3. Retrieval-stability summary
4. Statistical comparison results
5. Significant-result summary
6. Retrieval contamination figure
7. Baseline preservation figure

8. Answer-quality figure
9. Latency figure

The use of a fixed random seed and paired baseline comparisons allows the experiment to be reproduced under the same dataset, implementation, model configuration, and computational environment.

10. References

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The complete experimental implementation, including data preparation, retrieval evaluation, distractor-noise construction, statistical analysis, and result generation, is available in the accompanying Kaggle notebook:

Kaggle Notebook:

<https://www.kaggle.com/code/shaikjouziaafreenh/rag-retrieval-robustness-study>

The notebook provides the executable implementation used to reproduce the reported experimental results under the specified dataset and configuration.